Zixun Huang

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Research Interests

Theoretical foundations of machine learning, with an emphasis on **Optimization Theory & Scaling Laws**, **LLM Reasoning & Post-Training & Agentic RL**, and **applications of AI**.

Education

University of Pennsylvania, Wharton School Aug. 2026 – Present
Ph.D. in Statistics and Data Science

Peking University, School of Mathematical Sciences Sept. 2022 – Jun. 2026
B.S. in Statistics; Elite Undergraduate Program for Applied Mathematics; GPA: 3.7/4.0

University of California, Berkeley Jan. 2025 – Aug. 2025
Visiting Student; GPA: 4.0/4.0

Working Experience

Tencent HY – Qingyun Talent Program, *Research Intern* Jun. 2026 – Aug. 2026

Publications

[1] Jia-Nan Wang*, **Zixun Huang***, Kairui Li*, Lei Wu. *Momentum in Large-Batch Training: Polyak Enlarges the Critical Batch Size, Nesterov Improves Data Efficiency.* **Preprint**. [Paper]

[2] **Zixun Huang***, Kishan Panaganti*, Haitao Mi, Leowei Liang. *FlowSD: Trajectory-Balanced On-Policy Self-Distillation.* **Preprint**. [Blog]

[3] Zhongzhi Li*, Yucheng Shi*, Zongxia Li*, Ruhan Wang, Anhao Li, **Zixun Huang**, Junyao Yang, Lei Ke, Ninghao Liu, Haitao Mi, Leowei Liang. *Recursive Synthesis for Long-Horizon Terminal Tasks.* **Preprint**. [Paper] [Hugging Face] [Website]

[4] **Zixun Huang***, Jiayi Sheng*, Zeyu Zheng. *Variance-Aware Baselines and Adaptive Learning Rates for Reinforcement Learning with Verifiable Rewards.* **Under review**. [Paper]

[5] Yuhao Sun, Zekun Wu, **Zixun Huang**, Peijie Zhou. *TracingFlow: A Simulation-Free Trajectory Inference Framework Based on Second-Order Dynamics.* **Under review**. [Paper]

[6] Binghui Li*, Fengling Chen*, **Zixun Huang***, Lean Wang*, Lei Wu. *Functional Scaling Laws in Kernel Regression: Loss Dynamics and Learning Rate Schedules.* **NeurIPS 2025 Spotlight**. [Paper] * Equal contribution.

Research Experience

Optimization Theory & Scaling Laws

Peking University, *Research Intern*, Beijing May. 2023 – Aug. 2026
Supervisor: Assistant Professor Lei Wu

- **Large-batch momentum [1]:** Theoretically characterized how Polyak and Nesterov momentum reshape stochastic-training dynamics; established that Polyak enlarges the critical batch size, while Nesterov improves data efficiency.
- **Functional scaling laws [6]:** Developed an intrinsic-time analysis of one-pass SGD that predicts full loss trajectories under arbitrary learning-rate schedules; validated the theory on power-law kernels and 0.1B–1B language-model pre-training.

LLM Reasoning & Post-Training & Agentic RL

Tencent Hy LLM – Qingyun Talent Program, *Research Intern*, Beijing Jun. 2026 – Aug. 2026
Supervisor: Senior Researcher Kishan Panaganti

- **On-policy self-distillation [2]:** Developed FlowSD, a trajectory-balance-based OPSD algorithm for learning distributions over complete responses. Across five reasoning benchmarks, improved over GRPO by **1.95** and **2.12 points** on Qwen3-4B and Qwen3-8B, while increasing semantic strategy diversity among successful solutions.
- **Long-horizon agent training [3]:** Built a recursive synthesis pipeline that evolves verified terminal tasks into harder executable tasks, producing **37,484 tasks** across 15 rounds at approximately $0.05 per task; the resulting trajectories improved Qwen3.5 agents through SFT and RL.

University of California, Berkeley, *Research Intern*, Berkeley, CA Jan. 2025 – Aug. 2025
Supervisor: Associate Professor Zeyu Zheng

- **Variance-aware policy optimization [4]:** Derived a variance-optimal baseline and an SNR-adaptive learning-rate rule for KL-regularized RLVR with convergence guarantees. On Qwen3-4B-Base, OBLR-PO improved average Pass@1 over GRPO by **2.52 points** across four benchmarks.

Selected Honors and Awards

Grand Prize, Challenge Cup Competition	2026
Weiming Scholar, Peking University	2026
Hong Sheng Scholarship	2024
First Prize, Chinese Mathematics Contest	2024
Yau Contest Groups Prize	2024
Xiaomi Scholarship	2023
Gold Prize, Chinese Mathematical Olympiad	2021
Silver Prize, Chinese Mathematical Olympiad	2020

Skills

Programming: Python (NumPy, pandas, scikit-learn, PyTorch), C++, MATLAB, LaTeX
Language: GRE V160/Q170; GRE Math 970 (95th percentile); TOEFL 101